

A Firm-Level Microsimulation for VAT Policy Analysis^{*}

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Preliminary and Incomplete. Comments are welcome.

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Abstract

I present an open, reproducible firm-level microsimulation for costing reforms to the UK VAT registration threshold, released in full—including its synthetic data generator—as open-source code. The threshold is a *notch*: once turnover crosses £85,000, 20% VAT falls due on a firm’s entire turnover rather than only the excess, creating an incentive to bunch below. I build synthetic firm-level records from ONS business-structure counts (sector $\times$ turnover band $\times$ employment), calibrate them to HMRC VAT statistics for 2023–24, and use them to do three things on a common static basis. First, I cost threshold-location reforms and schedule reforms (a graduated taper and a reduced-rate band) statically; the static costing reproduces HMRC’s published figure for the £85,000 to £90,000 reform (–£175m versus –£185m in 2025–26). Second, I document firm bunching at the £85,000 notch, estimating a bunching ratio of $b \approx 0.060$, consistent with [Liu et al. \(2021\)](#) once normalised; a bunching-implied turnover elasticity of ≈ 0.45 follows from the notch-width relation, which I report descriptively and with caveats. Third, I characterise the notch’s distortion through its *dominated region*—the exact, model-free range of turnover above the threshold in which no firm locates—which is £21,250 wide (£85,000–£106,250), and show how each reform changes it. I treat the model as a turnover-tax-notch approximation for consumer-facing, limited-reclaim firms, abstracting from input reclaim and voluntary registration; here the notch model supplies the exact dominated region rather than a behavioural pricing engine. For each reform I report the static cost together with the portion of the dominated region it removes: raising the threshold to £100k (–£374m) relocates the £21,250 zone without

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removing it; a banded 10% reduced rate (−£357m) shrinks it to £9,444; a 15% reduced rate (−£178m) shrinks it to £15,000; and a graduated taper over [£85k, £105k] (−£335m) removes it. Behavioural firm re-optimisation under counterfactual schedules is left to future work.

Keywords: value-added tax, microsimulation, tax notch, policy

1 Introduction

The United Kingdom’s Value Added Tax (VAT) applies above a registration threshold that is among the highest in the OECD (Seely, 2024; HM Revenue and Customs, 2024). The threshold is a *notch* rather than a kink: once a firm’s annual taxable turnover crosses it, the standard 20% rate falls due on all of the firm’s turnover, not only on the part above the threshold, so registration produces a discrete jump in liability rather than a marginal one. The threshold falls in a dense part of the firm distribution. Figure 1(a) shows that UK firms are concentrated at the lower end of the turnover distribution, and Figure 1(b) shows that VAT-registered firms are concentrated in the turnover bands at and below the threshold, so the notch applies to a large number of firms.

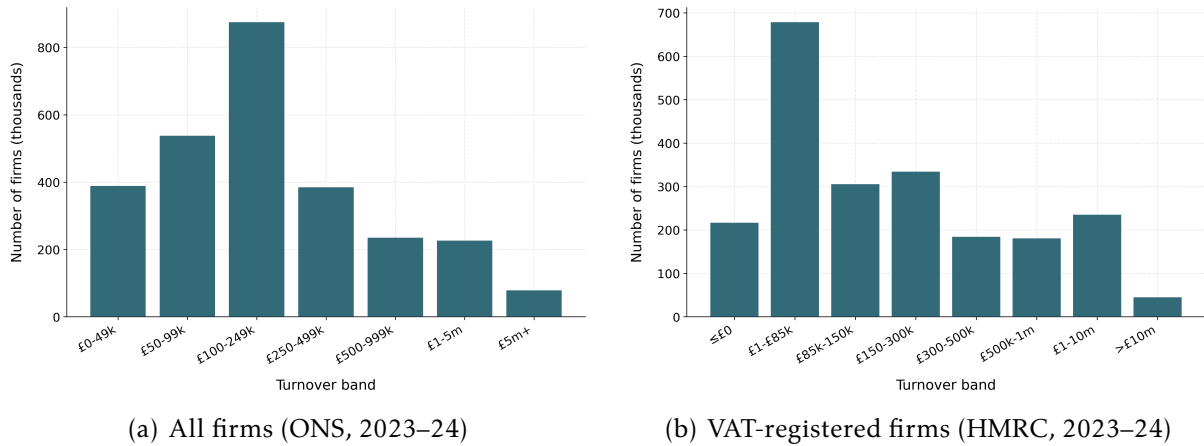


Figure 1: The UK firm-size distribution and VAT registration, 2023-24 (threshold £85,000). Panel (a): number of firms by turnover band. Panel (b): number of VAT-registered firms by turnover band. Source: panel (a), ONS, *UK Business: Activity, Size and Location* (firm counts by turnover band); panel (b), HMRC, *Value Added Tax (VAT) Annual Statistics* (VAT-registered population by turnover band).

Firms respond to the notch by locating below the threshold. Tabulations of HMRC turnover data show the number of firms rising towards £84,000 and then dropping sharply at the £85,000 threshold, leaving excess mass just below it (Office for Budget Responsibility, 2023). Liu et al. (2021) document this bunching in UK firm data, and Liu et al. (2024) find that firms slow their turnover growth as they approach the threshold. The notch therefore distorts the size distribution and the growth decisions of firms near the margin, which places the threshold under recurrent pressure for reform.

Reform proposals change the threshold in one of three ways: its *level* (raising or lowering it), its *shape* (replacing the notch with a schedule that phases the rate in over a band), or its *rate* (a reduced rate for firms in a band above the threshold). The three are not equivalent. Changing the level moves the notch along the turnover axis; changing the shape or the rate alters the size of the discrete jump itself. Costing the three on a

common basis is the practical difficulty this paper addresses. A change in the level can be costed by mechanical accounting, but a change in the shape of the schedule cannot be evaluated with the reduced-form bunching methods used to study the threshold, because a continuous schedule leaves no single point of excess mass to exploit. The firm-level data needed to study turnover near the threshold is, in addition, not available in open form.

In this paper I build an open, reproducible firm-level microsimulation for costing UK VAT threshold reforms on a common static basis. I construct synthetic firm microdata calibrated to published HMRC aggregates—band-level firm counts and sector turnover totals—for the 2023–24 tax year, when the threshold was £85,000 (raised to £90,000 in April 2024); the synthetic data can be released openly and regenerated under any counterfactual schedule. On these data I do three things. I cost both level reforms and schedule reforms—a graduated taper and a reduced-rate band—statically, by applying each counterfactual schedule mechanically to the firm distribution. I measure firm bunching at the notch with a reduced-form bunching estimator in the tradition of [Saez \(2010\)](#) and [Kleven and Waseem \(2013\)](#), and report a bunching-implied turnover elasticity descriptively and with caveats. And I characterise the notch’s distortion through its *dominated region*—the exact, model-free range of turnover above the threshold in which no firm locates—and how each reform changes it. The behavioural re-optimisation of each firm under a counterfactual schedule—a forward solve that re-prices firms rather than applying the schedule to the observed distribution—is left to future work.

Three findings follow. First, the behavioural response visible at the threshold is modest. I estimate a bunching ratio of $b \approx 0.060$, consistent with the bunching documented by [Liu et al. \(2021\)](#) once normalised. The notch-width relation implies a turnover elasticity of about 0.45, which I report descriptively and with caveats: it is a static, contemporaneous statement about excess mass near the threshold, sensitive to the excluded-window choice, and is separate from the dynamic growth-suppression channel ([Liu et al., 2024](#); [Bashir et al., 2024](#)). Because the response is small relative to the mass affected, the static revenue cost of changing the threshold’s level is close to its mechanical value: my static costing of the April 2024 increase from £85,000 to £90,000 gives £175m in 2025–26 against HMRC’s published £185m, and forward moves from the £90,000 base range from a gain of £665m at £70,000 to a cost of £1,205m at £120,000, against a base of about £203.8bn.

Second, the shape and rate reforms act on the notch’s distortion in a way that a change in level does not. The notch creates a *dominated region* of width $a = T^*\tau/(1-\tau) = £21,250$ above the threshold (£85,000–£106,250), a range of turnover in which no firm can locate without being strictly worse off than at the threshold. This region is an exact, model-free property of the schedule: it follows from the arithmetic of the notch alone and requires no behavioural calibration. Raising the threshold relocates this region without removing it. A reduced rate shrinks it—to £9,444 at a 10% band rate and £15,000 at 15%—while a graduated taper, being continuous, removes it. Third, the reforms have comparable static revenue costs: £374m to raise the threshold to £100,000, £335m for the taper, and £357m and £178m for the 10% and 15% reduced rates. I report, for each reform, this

static cost together with the portion of the dominated region it removes.

Related literature. The paper draws on and contributes to three strands of work. The first is the methodology of bunching and notches (Saez, 2010; Kleven and Waseem, 2013; Chetty et al., 2011; Kleven, 2016; Best et al., 2015), which provides the reduced-form estimator and the analytic notion of a dominated region; I apply the estimator descriptively and use the dominated region to go beyond the reduced form, characterising a distortion that leaves no single point of excess mass to exploit. The second is the empirical literature on the VAT registration threshold (Liu et al., 2021, 2024; Onji, 2009; Harju et al., 2019; Asatryan and Peichl, 2017; Nandi and Warwick, 2020; Bashir et al., 2024), which estimates the behavioural response and studies changes in the threshold’s level. Liu et al. (2021) is closest: it estimates bunching at the UK notch on administrative data and documents voluntary registration; I report a consistent bunching estimate on open synthetic data and a caveated bunching-implied turnover elasticity, and I cost shape and rate reforms statically on a common basis. The international evidence (Harju et al., 2019; Onji, 2009; Asatryan and Peichl, 2017; Nandi and Warwick, 2020) notes that bunching at a VAT threshold can reflect compliance costs, firm-splitting, or reporting rather than real turnover responses. The third strand is the structural literature on size-based thresholds (Garicano et al., 2016; Gourio and Roys, 2014), which identifies policy-invariant primitives at the French 50-employee threshold and computes welfare from them; the present paper makes no structural claim and instead prices reforms statically. This paper makes two contributions. First, it provides an open, reproducible firm-level microsimulation for costing UK VAT threshold reforms, where the existing firm-level evidence on the threshold rests on restricted administrative data. Second, it characterises the notch’s distortion through its exact, model-free dominated region and costs two schedule reforms—a graduated taper and a reduced-rate band—on the same static basis as level reforms, where prior work has neither costed these reforms nor measured this distortion. The behavioural re-optimisation of firms under counterfactual schedules, a structural or dynamic turnover elasticity, and a welfare and marginal-value-of-public-funds accounting—together with revenue-neutral designs and the treatment of voluntary registration—are left to future work.

The remainder of the paper proceeds as follows. Section 2 describes the institutional background of the UK VAT threshold. Section 3 sets out the synthetic firm microdata and its calibration to HMRC aggregates. Section 4 sets out the static costing of threshold changes and the schedule reforms. Section 5 reports the reduced-form bunching evidence. Section 6 develops the notch and its exact dominated region, and shows how each reform changes it. Section 7 concludes.

2 Institutional background

Registration and the notch. The United Kingdom levies a value-added tax at a standard rate of 20% on most goods and services. A business must register once its taxable turnover over any rolling twelve-month period exceeds the registration threshold, after

which it charges VAT on its taxable sales and reclaims VAT paid on inputs. Because registration makes VAT due on the firm's *entire* taxable turnover rather than only the amount above the cutoff, the threshold is a *notch*—a discrete jump in liability—and not a kink. The cost of crossing is largest for consumer-facing firms with limited pass-through and small reclaimable input shares, for whom the incentive is to restrain turnover and locate just below the threshold (Liu et al., 2021; Kleven and Waseem, 2013).

Value added, input reclaim, and voluntary registration. Real VAT is a tax on *value added*, not on turnover: a registered firm charges output VAT on its sales but reclaims input VAT on its purchases. A large share of below-threshold firms therefore register *voluntarily*—roughly 45% of eligible UK firms (Liu et al., 2021), driven by input-VAT reclaim and business-to-business sales—and for them the incentive to bunch is weak. My one-input model taxes *turnover* rather than value added, so it is a *turnover-tax-notch approximation* for the consumer-facing, limited-reclaim firms for whom the bunching incentive is sharpest; it abstracts from input reclaim and voluntary registration. I treat this as a first-order scope restriction.

Threshold history. The registration threshold was held at £85,000 from 1 April 2017 to 31 March 2024—a seven-year nominal freeze, the longest in the history of the tax—before being raised to £90,000 on 1 April 2024, with the deregistration threshold rising from £83,000 to £88,000 (HM Revenue and Customs, 2024; Seely, 2024). Because nominal turnover grows with inflation and real activity, freezing the threshold lowers it in real terms and draws an increasing number of firms towards the registration margin; this fiscal-drag mechanism was a central motivation for the April 2024 increase. My data year is 2023–24, so the prevailing statutory threshold throughout my analysis is £85,000. Figure 2 shows the pattern in the turnover distribution: the number of firms rises towards £84,000 and falls sharply above £85,000, and the projected 2025–26 distribution is more sharply bunched than the 2019–20 outturn as fiscal drag draws more firms towards the frozen threshold.

The reform debate. The UK threshold is among the highest in the OECD, and there is debate over whether to raise it—to reduce the compliance burden on small firms—or lower it, to broaden the base and reduce the growth distortion at the margin. Three families of reform feature, and I evaluate all three: a change in the threshold's *level* (the £85,000 to £90,000 increase of April 2024 and further moves around it); a change in its *shape* (a *graduated taper* that phases the effective rate in over a band, replacing the notch with a continuous profile); and a change in its *rate* (a *reduced rate* for small firms in a band above the threshold, with the standard 20% rate retained above). A banded reduced rate introduces a smaller *secondary* notch at the band top, which the continuous taper avoids. I cost all three by the same machinery in Section 4.

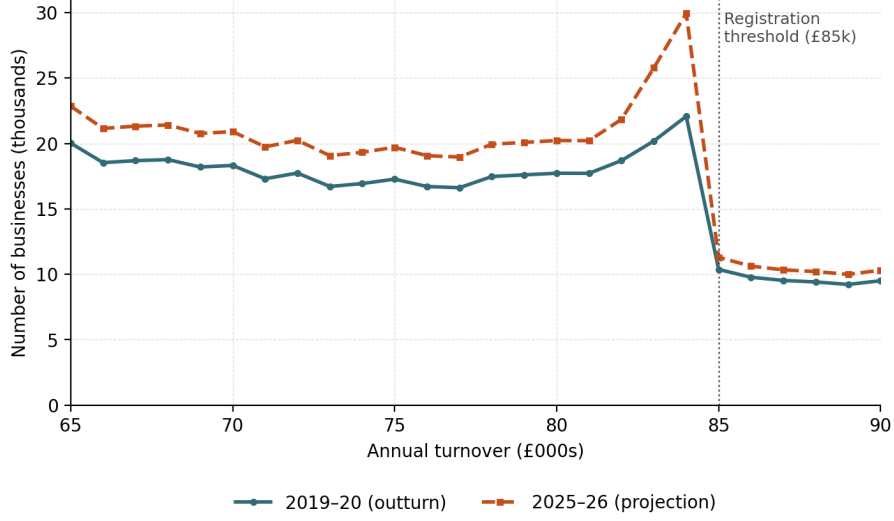


Figure 2: Bunching below the £85,000 VAT registration threshold: number of businesses by £1,000 turnover band. Source: [OBR, *Economic and Fiscal Outlook March 2023*](#), Chart C (Open Government Licence).

3 Data

Studying firm responses at the threshold needs firm-level turnover near the cutoff, which the official statistics publish only in coarse turnover bands. I therefore build a synthetic firm population calibrated to those aggregates: it reproduces the official totals while resolving the individual firm that the authorities do not release, opening the analysis to anyone. The same calibrate-to-published-aggregates approach underlies open household-level tax-benefit microsimulation ([PolicyEngine, 2025](#)); here I apply it at the firm level. The generator takes the registration threshold as a parameter and is documented in [Appendix A.1](#).

Construction. Index firms by i . For each sector s and ONS turnover band $b = [\underline{b}, \bar{b}]$, the *UK Business: Activity, Size and Location* table gives a firm count $N_{s,b}$, and I draw $N_{s,b}$ firms with within-band turnover

$$y_i = \underline{b} + (\bar{b} - \underline{b}) u_i + \varepsilon_i, \quad u_i \sim \mathcal{U}(0, 1), \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_b^2),$$

truncated below at zero, where ε_i smooths the within-band distribution. Each firm is given input expenditure $x_i = \rho_i y_i$, with the input-output ratio ρ_i drawn from a rescaled Beta distribution with sector-specific shifts and bounded to $\rho_i \in [0.6, 1.5]$; its value added, the net VAT base, is $v_i = y_i - x_i$. Employment is assigned from the ONS employment-band shares.

The base population reproduces the ONS structure but not the HMRC VAT-registered totals, so I re-weight it. Each firm receives a strictly positive weight $w_i = e^{\theta_i}$, and for a

vector of official targets $T = (T_k)_{k=1}^K$ the weighted synthetic counterpart of target k is

$$\hat{T}_k(\theta) = \sum_i a_{ki} w_i,$$

where a_{ki} is firm i 's contribution to target k : an indicator of band or sector membership for a count target, and the firm's liability v_i for a VAT-liability target. I choose θ to minimise a symmetric relative-error loss with per-target importance weights λ_k ,

$$L(\theta) = \frac{1}{K} \sum_k \lambda_k \min \left\{ \left(\hat{T}_k / T_k - 1 \right)^2, \left(T_k / \hat{T}_k - 1 \right)^2 \right\} + \alpha \sum_i |\theta_i|,$$

the final term an L_1 penalty on the log-weights. The targets are the HMRC VAT-registered counts by turnover band and by sector, the HMRC net VAT liability by band and by sector, and the ONS employment-band totals; turnover-band targets carry the largest λ_k , followed by VAT liability by band. The problem is solved by gradient descent, and the symmetric loss keeps targets of very different magnitudes on a common scale.

A firm is registered with certainty when its turnover exceeds the threshold T^* . Below the threshold it registers voluntarily with probability $r = M/n_<$, the HMRC count of registered firms in the £1-threshold band, M , divided by the synthetic number of below-threshold firms, $n_<$, so the synthetic voluntary-registration rate matches the HMRC aggregate. The resulting dataset comprises approximately 2.94 million firm records, weighted to represent roughly 2.0 million UK firms; validated against the official sources along every calibrated dimension, it attains an overall calibration accuracy of about 87%, reproducing the aggregates closely while supplying the firm-level resolution they lack.

The registration threshold in the data. My data are for the 2023–24 year, in which the statutory threshold was £85,000. The firm density steps down at the threshold—from roughly 10,400 to 6,700 firms per £1,000 turnover band—reflecting the incentive to remain below the registration cutoff, with a second, smaller step at £150,000, the ceiling of the VAT Flat Rate Scheme (Figure 3). Counterfactual policies—including the £85,000 to £90,000 reform and the forward threshold sweep of Section 4—are evaluated on these firms, by re-applying the registration rule to the microdata for the static costings and by re-solving firm optimisation for the behavioural exercise developed in a companion structural step. I hold the empirical turnover distribution fixed and let behaviour respond to the reform, rather than imposing a bunching pattern at the counterfactual threshold, which would presuppose the response the analysis sets out to measure.

4 Static costing of threshold reforms

A static costing measures the first-order revenue effect of a reform by re-applying the tax rule to a fixed firm population, with no behavioural response. Let firm i have turnover y_i , calibrated net VAT liability v_i (output less input VAT; Section 3), and weight w_i . A firm remits VAT only once registered, that is once its turnover exceeds the threshold T ,

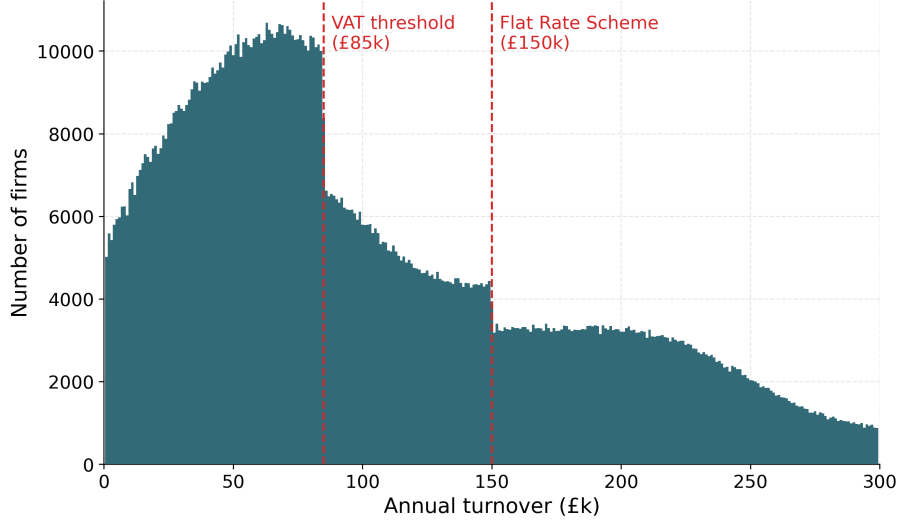


Figure 3: Synthetic firm turnover distribution (weighted firm counts, £1,000 bins), 2023–24. The density steps down at the £85,000 registration threshold and again at the £150,000 Flat Rate Scheme ceiling.

so total VAT revenue at threshold T is the weighted sum of liabilities over registered firms,

$$R(T) = \sum_i w_i v_i \mathbf{1}\{y_i \geq T\}.$$

Because turnover is held fixed, moving the threshold from T_0 to T changes revenue only through the firms whose registration status flips—those with turnover in the band between the two thresholds:

$$\Delta R = R(T) - R(T_0) = -\text{sign}(T - T_0) \sum_i w_i v_i \mathbf{1}\{T_{\min} \leq y_i < T_{\max}\},$$

where $T_{\min} = \min(T, T_0)$ and $T_{\max} = \max(T, T_0)$: lowering the threshold draws firms into the net and raises revenue, while raising it releases firms and loses revenue. The estimate is therefore the mechanical reclassification effect and excludes the behavioural response of Section 5. The calibrated weights reproduce HMRC’s net VAT-liability totals (Section 3), and I age the microdata to a given fiscal year by a cumulative nominal-growth factor. At the £90,000 threshold this puts total VAT revenue at £203.8bn in 2025–26 and £210.0bn in 2026–27.

Validation against HMRC. I first check the static pipeline against HMRC’s own costing of the April 2024 increase from £85,000 to £90,000. Because my microdata are calibrated to HMRC VAT aggregates (Section 3), a close match confirms internal consistency rather than out-of-sample accuracy. Figure 4 compares the two costings year by year: they share sign and profile throughout—a loss that narrows as fiscal drag lifts the frozen baseline towards the reformed threshold, turning to a small gain by 2028–29—and agree

to within £7m–£21m per year (for 2025–26, –£175m against HMRC’s –£185m). The model reproduces both the magnitude and the turning point of the official costing.

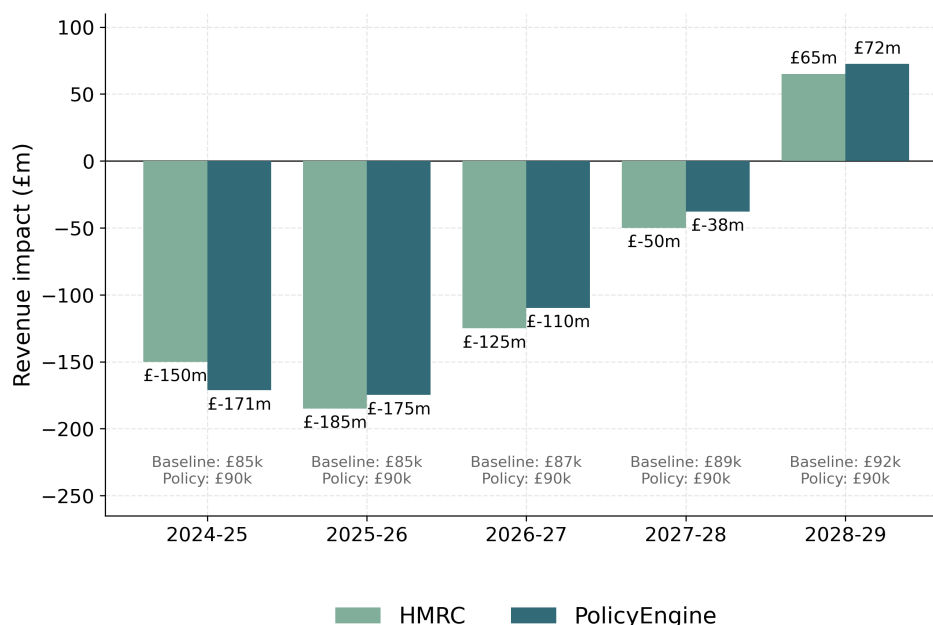
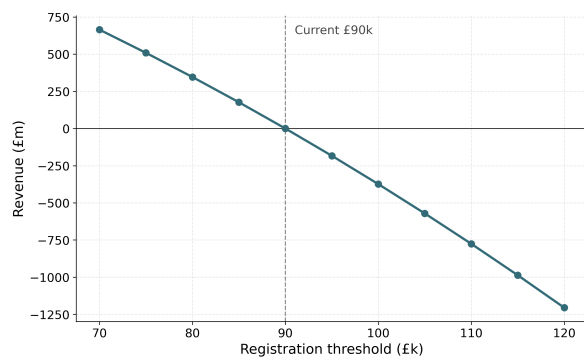


Figure 4: Anchor reform: revenue impact of raising the VAT registration threshold from £85,000 to £90,000 (the April 2024 change), my estimate versus HMRC’s published costing, 2024–25 to 2028–29. The two series share sign and profile, both turning positive in 2028–29 as fiscal drag lifts the frozen baseline above the reformed threshold. Because the microdata are calibrated to HMRC aggregates, this is a consistency check, not an out-of-sample validation.

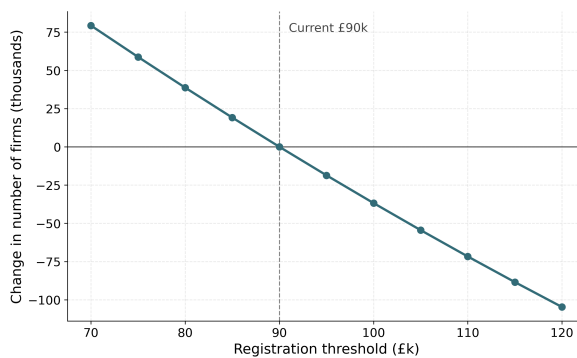
A menu of threshold locations. Having checked the method, I use it to cost a menu of further threshold locations, taking the post-reform £90,000 threshold as the baseline. Each row of Table 1 costs a move of the threshold to a new location for 2025–26, holding the turnover distribution fixed, so the revenue change reflects only the reclassification of firms into or out of the VAT net. Because the calibration places extra weight on high-liability firms just below the threshold, I evaluate the reclassified band from a smooth extrapolation of the clean above-threshold profile rather than the raw below-threshold mass. Lowering the threshold draws firms in and raises revenue; raising it loses revenue. The schedule is close to linear—each £5,000 step changes revenue by roughly £150m–£220m, with the marginal effect growing as the threshold sweeps across denser parts of the distribution. Even a £30,000 rise to £120,000 costs about £1.2bn, under 0.6% of the £203.8bn base, because the threshold governs many small firms whose individual liabilities are small. Figure 5 shows the sweep graphically. These are *static* results: they bound the first-order fiscal effect of a threshold change but abstract from the behavioural response of Section 5.

Table 1: Static revenue and VAT-paying-firm effects of moving the VAT registration threshold, 2025–26, relative to the post-reform £90,000 baseline (total VAT revenue £203.8bn). Figures are mechanical, holding firm turnover fixed (no behavioural response). This sweep is distinct from the anchor reform of Figure 4.

Threshold	Revenue change (£m)	Change (%)	VAT-paying firms (000s)
£70,000	+664.9	+0.33	+79.3
£75,000	+509.0	+0.25	+58.7
£80,000	+346.3	+0.17	+38.7
£85,000	+176.6	+0.09	+19.1
£90,000 (baseline)	0	0	0
£95,000	−183.5	−0.09	−18.6
£100,000	−374.0	−0.18	−36.8
£105,000	−571.4	−0.28	−54.5
£110,000	−775.7	−0.38	−71.7
£115,000	−987.0	−0.48	−88.5
£120,000	−1,205.1	−0.59	−104.7



(a) Revenue impact (£m)



(b) Change in VAT-paying firms (000s)

Figure 5: Static effect of moving the VAT registration threshold, 2025–26, relative to the £90,000 baseline (dashed line). Panel (a): revenue impact—lowering the threshold raises revenue, raising it loses revenue, and the schedule is approximately linear over the £70,000–£120,000 range. Panel (b): the change in the number of VAT-paying firms, tracking the local density of firms on each side of the threshold.

4.1 Costing schedule reforms

The threshold sweep above moves the *location* of a single cutoff. The same static method—re-applying a tax rule to the fixed turnover distribution, with no behavioural response—also costs reforms that change the *shape* of the schedule rather than merely its position. I consider three reform types and report their first-order revenue effects against the £85,000-notch baseline (VAT base £179.3bn for the schedule reforms; the threshold move is measured against the post-reform £90,000 baseline of Table 1). For each reform I recompute the weighted sum of liabilities $R = \sum_i w_i v_i$ under the new schedule, holding each firm’s turnover fixed, and difference it against the baseline.

Threshold relocation. Raising the threshold to £100,000 is a pure location move, costed exactly as in the sweep above: it releases the firms with turnover in the £90,000–£100,000 band from the VAT net, for a static revenue change of –£374m (Table 1).

Graduated taper. A graduated taper replaces the discrete notch with an effective rate that phases linearly from 0% at £85,000 to the full 20% at £105,000. Firms in the band remit a tapered rather than a full rate on their (fixed) turnover; mechanically this reduces revenue by –£335m, with about 125,600 firms in the £85,000–£105,000 band facing a lower effective rate.

Banded reduced rate. A banded reduced rate charges firms in the £85,000–£105,000 band a reduced rate $\tau_{\text{low}} \in \{10\%, 15\%\}$ in place of the standard 20%, with 20% retained above £105,000. Statically this reduces revenue by –£357m at $\tau_{\text{low}} = 10\%$ and by –£178m at $\tau_{\text{low}} = 15\%$, again with about 125,600 firms in the band facing the lower rate.

Table 2: Static, behaviour-free revenue effect of the three reform types relative to the £85,000-notch baseline (VAT base £179.3bn for the schedule reforms; the threshold move is measured against the post-reform £90,000 baseline of Table 1). Each reform is costed mechanically by re-applying its schedule to the fixed turnover distribution, with no behavioural response.

Reform	Lever	Static revenue (£m)
Raise threshold to £100,000	Location	–374
Graduated taper [£85k, £105k]	Shape (phase-in)	–335
Reduced rate 10% [£85k, £105k]	Rate (step)	–357
Reduced rate 15% [£85k, £105k]	Rate (step)	–178

These static figures bound the first-order fiscal effect of each reform but abstract from the behavioural response of Section 5. One feature of the shape reforms is not visible in a static costing: unlike a threshold move, which simply transports the notch to a new cutoff, the taper and the reduced rate *change the size of the notch itself*—the discrete drop in net revenue at the threshold and the empty band of turnover it induces. Quantifying that change requires the structural object I develop in Section 6, where I

derive the notch’s dominated region and show analytically how each of these reforms alters it.

5 Bunching at the VAT Threshold

This section presents reduced-form evidence of firm bunching at the VAT registration threshold, using the polynomial counterfactual-density estimator of [Saez \(2010\)](#) and [Kleven and Waseem \(2013\)](#) as applied to UK VAT data by [Liu et al. \(2021\)](#); I also draw on the synthesis of [Kleven \(2016\)](#) and the friction diagnostics of [Chetty et al. \(2011\)](#). The estimator is purely reduced-form: it fits a smooth counterfactual to the empirical turnover density and reads excess mass off the gap, without imposing any structural model of the firm. The notch mechanism that generates this mass—VAT applying to a registered firm’s entire turnover, so that locating just below the cutoff is attractive—is the subject of Section 6; here it suffices that the notch produces excess mass below the threshold and missing mass above it. I estimate bunching at the £85,000 threshold prevailing in my 2023–24 data.

5.1 Methodology

Following [Liu et al. \(2021\)](#), I group firms into fine turnover bins of £100 and estimate a counterfactual density—the distribution that would obtain absent the registration notch—by polynomial regression on bin counts, excluding a manipulation window around the threshold y^* :

$$c_j = \sum_{l=0}^q \beta_l (y_j)^l + \sum_{i \in [y_-^*, y_+^*]} \gamma_i 1\{j = i\} + \varepsilon_j,$$

where c_j is the number of firms in bin j , y_j is the distance of bin j from the threshold, q is the polynomial order, and $[y_-^*, y_+^*]$ is the excluded window. The fitted polynomial is then projected into the excluded region to recover the counterfactual density $g(y)$. Excess bunching is the integrated difference between the observed and counterfactual densities just below the threshold,

$$B(y^*) = \int_{y^* - \Delta y^*}^{y^*} g(y) dy \approx g(y^*) \Delta y^*,$$

the approximation holding when g is locally smooth. The bunching ratio $b(y^*) = B(y^*)/g(y^*) \approx \Delta y^*/y^*$ is the width of the bunching region expressed as a fraction of the threshold—a normalised *ratio*, not an elasticity. Translating the excess mass into a behavioural elasticity requires a further set of assumptions, discussed with explicit caveats in the comparison below; the estimator here delivers the raw geometry of the response.

5.2 Results

For the 2023–24 data, measured at the £85,000 threshold, my *preferred* specification imposes the mass-conservation restriction that excess mass below the threshold equal missing mass above it, locates the marginal buncher endogenously, and bootstraps the sampling distribution (Appendix A.2). It yields a bunching ratio $b = 0.060$ (SE 0.005) and excess mass of 8,712 firms (SE 662), reproducible by `python -m bunching` on the regenerated 2023–24 synthetic data; full inference (bootstrap standard errors and the mass-conservation constraint) is reported in Appendix A.2. Mass conservation matters: an unconstrained polynomial fit implies excess against missing mass that grossly violate adding-up, whereas the constrained estimator reconciles the two. In all specifications the observed density displays clear excess mass in the £100 bins immediately below the threshold and a corresponding deficit just above it, the signature of a notch-induced behavioural response. A non-trivial share of firms near the threshold thus adjust their turnover to avoid registration.

It is worth fixing at the outset that the bunching evidence speaks to *several* distinct elasticity-like objects, which the literature too often runs together and which I keep separate throughout. (i) The *notch-width elasticity* (≈ 0.45 , Method B below) infers a turnover elasticity from the empirical width of the excess-mass span in the Kleven–Waseem manner; it is the bunching-implied elasticity I report descriptively in this section. (ii) The *marginal-buncher elasticity* (≈ 0.66) reads the indifference condition of the firm at the upper edge of the bunching segment. (iii) The CES *substitution elasticity* $\sigma \approx 4.85$ is a wedge-substitution parameter recovered from the above/below mass ratio, not a turnover-supply elasticity at all. All three are bunching-based geometric readings of the same excess mass; none should be read as a substitute for another, and—for the reasons set out below—none should be read as a welfare-sufficient structural turnover elasticity.

5.3 Comparison with the literature

The closest comparator is Liu et al. (2021) (hereafter LLAT), who estimate bunching at the same £85,000 UK VAT notch over a manipulation window of $-\text{£}14\text{k}$ to $+\text{£}24\text{k}$. To compare like with like I normalise my excess mass to the LLAT-style excess-bunching statistic (excess mass per unit of counterfactual density at the threshold), which yields $b_{\text{LLAT}} = 0.94$ (SE 0.07, 95% CI [0.80, 1.09]) against their $b = 1.361$ (SE 0.202). The two are of the same order of magnitude; my synthetic-data response is somewhat less pronounced. This is what one would expect: my firm records are built up from ONS business-structure counts and calibrated to aggregate HMRC totals rather than drawn from the administrative micro-data LLAT observe, so the near-threshold spike is smoothed by the synthesis step and by survey weighting, and the bootstrap re-binning further attenuates the peak. The gap is therefore a feature of the data-generating process, not evidence of a different behavioural mechanism. Like LLAT, my setting features substantial voluntary registration below the threshold—LLAT report that roughly 29.8% of firms

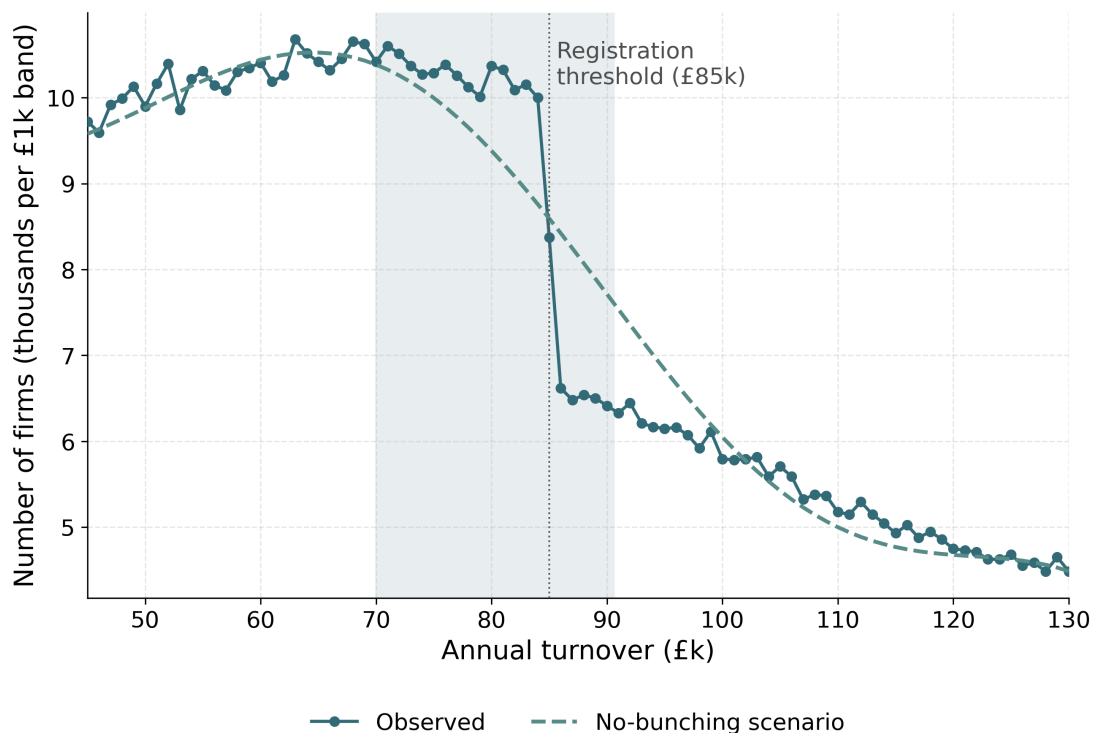


Figure 6: Bunching analysis at the £85,000 threshold (2023–24 data year). The figure shows the observed firm density, the estimated polynomial counterfactual density fitted outside the manipulation window, and the implied distribution under a forward-mapped £95k new-policy threshold. The headline estimates, from the mass-conserving estimator with bootstrap inference (Appendix A.2) and reproducible by `python -m bunching` on the regenerated 2023–24 synthetic data, are a bunching ratio $b = 0.060$ (a width ratio, not an elasticity), excess mass $E = 8,712$ firms, and substitution elasticity $\sigma = 4.85$.

register voluntarily—which dampens observed bunching relative to a pure turnover-tax notch and which my turnover-tax-notch model abstracts from.

Several caveats govern the elasticity comparison, and I emphasise that the bunching-implied turnover elasticity (≈ 0.45 , Method B, from the notch width) is a *descriptive* reading of the excess-mass geometry, not a structural estimate I stand behind. First, LLAT *deliberately decline* to report a turnover or output elasticity, arguing that it is not point-identified at a notch and is not a welfare-sufficient statistic; their excess-bunching ratio is the object they stand behind, and I follow them in not treating my number as a sufficient statistic for welfare. Second, the bunching-implied number is attenuated and confounded by optimisation frictions: observed bunching is far smaller than a friction-less notch model predicts, so any elasticity backed out of the raw excess mass mixes the structural response with friction. Third, a turnover notch confounds an output-supply response with a demand-side (pass-through) response, so the bunching-implied number is an upper envelope on the pure supply elasticity rather than a clean structural elasticity. My bunching-implied elasticities—0.45 from the notch width (Method B) and 0.66 from the marginal-buncher indifference condition—should therefore not be read as comparable to an LLAT estimate, but against the Kleven–Waseem structural-elasticity tradition (Kleven and Waseem, 2013), whose notch-derived taxable-income elasticities fall below ~ 0.3 once optimisation frictions are allowed for. My estimates sit somewhat above that band, consistent with the supply-plus-pass-through confound just noted. A close methodological analogue is Almunia and Lopez-Rodriguez (2018), who find Spanish firms at an enforcement notch suppress revenue by up to 7.5%. The OBR figures are corroborating order-of-magnitude context, not a like-for-like match: Office for Budget Responsibility (2023) count firms holding turnover below the frozen threshold over a *wide* band (rising from 23,000 in 2017–18 to $\sim 44,000$ in 2025–26, with foregone turnover of £110m rising to £350m), a different object from my excess-mass spike of $\sim 8,700$ firms in the bins immediately below the cutoff.

5.4 From reduced-form evidence to structure

These estimates quantify the aggregate response at the prevailing threshold but do not identify the underlying firm-level productivity distribution. For a move of the threshold itself the reduced-form counterfactual already suffices; what it cannot price is a change to the *shape* of the schedule, which has no single point of excess mass to forward-map. What the structure of Section 6 *does* deliver exactly, for every reform considered, is the notch’s dominated region and how each schedule change alters it; a fully behavioural forward solve that re-prices each firm’s turnover under counterfactual schedules would require further parameter-sensitive assumptions and is left to future work.

6 The VAT Notch and Its Dominated Region

The reduced-form evidence in Section 5 establishes that firms respond to the VAT registration notch. To characterise that notch precisely I set out its firm-level structure, following the framework of Kleven and Waseem (2013), Saez (2010), and Kleven (2016), as applied to UK VAT by Liu et al. (2021). The UK registration threshold is a *notch*, not a kink: once turnover crosses the threshold the firm becomes liable for VAT on its *entire* turnover, not merely on the increment above it. I model this notch exactly. The section’s central deliverable is the notch’s *dominated region*—the exact, model-free measure of the distortion the threshold imposes (Section 6.2), whose analytic response to each schedule reform complements the static reform costs of Section 4.1. The dominated region is the robust, headline structural object: it depends on the rate and threshold alone and is therefore exact. A behavioural re-optimisation of firms under counterfactual schedules—a forward solve that prices the response in turnover to reforms the data never observe—would require additional, parameter-sensitive assumptions about the firm’s technology and elasticity, and I leave that dynamic layer to future work.

Scope: a turnover-tax notch. The structure here taxes *turnover*. Real VAT is levied on *value added* with input-tax reclaim, and a substantial share of below-threshold firms register voluntarily to recover input VAT (Liu et al., 2021). My treatment abstracts from input reclaim and voluntary registration: it is a turnover-tax-notch approximation. This is a scope limitation I state up front; it bounds the welfare claims I make and is one I return to in the concluding discussion.

6.1 The firm’s problem with a VAT notch

Each firm i chooses turnover y_i to maximise after-tax profit. Let $c_i(y_i)$ denote the cost of producing turnover y_i , T^* the registration threshold, and τ the standard VAT rate. Because VAT applies to the firm’s entire turnover once it registers, after-tax revenue is discontinuous and profit is

$$\pi_i(y_i) = \begin{cases} y_i - c_i(y_i), & y_i < T^* \quad (\text{unregistered}), \\ (1 - \tau) y_i - c_i(y_i), & y_i \geq T^* \quad (\text{registered}). \end{cases} \quad (1)$$

At T^* after-tax revenue falls discretely by τT^* : a firm that nudges its turnover from just below to just above the threshold loses τT^* of net revenue on turnover it was already earning. With the standard rate $\tau = 0.20$ and threshold $T^* = \text{£}85,000$ this discrete drop is $\tau T^* = \text{£}17,000$. This drop—absent from a kink, where only the *marginal* rate changes at the threshold while average liability moves continuously—is the defining feature of a notch and the source of the bunching documented in Section 5. It is what makes locating just below the cutoff strictly attractive and creates the empty band derived in Section 6.2.

6.2 The dominated region

The notch creates a range of turnover just above T^* that *no* firm chooses, because every point in it yields strictly lower profit than locating at T^* . A firm at the threshold earns net revenue T^* ; a registered firm at $T^* + a$ earns $(1 - \tau)(T^* + a)$. Equating the two—the cost difference across this narrow band is negligible—the upper edge of the dominated region is

$$(1 - \tau)(T^* + a) = T^* \implies a = T^* \frac{\tau}{1 - \tau}. \quad (2)$$

This is the Kleven–Waseem dominated-region width (Kleven and Waseem, 2013). At my parameters it gives $a = £85,000 \times 0.20/0.80 = £21,250$, so the dominated region is $(T^*, T^* + a) = (£85,000, £106,250)$. Two properties make this $\sim £21\text{k}$ band the load-bearing analytic object of the paper. First, it is *exact*: it depends on τ and T^* alone and is independent of the cost specification, of any behavioural elasticity, and of any smoothing parameter. Second, it is an order of magnitude wider than the $\sim £1\text{k}$ region implied by a smoothed *kink* approximation. The dominated region is the structural counterpart of the missing mass measured just above the threshold in Section 5: the reduced-form deficit and the model-free empty band are the same object viewed two ways. The contrast with a smoothed-kink approximation, and a robustness check that a smooth schedule reproduces this width, are taken up in a separate robustness appendix.

How each reform changes the dominated region. Because the width $a = T^*\tau/(1 - \tau)$ depends on the threshold and the rate alone, the effect of each schedule reform of Section 4.1 on this misallocation zone is exact and follows directly from (2), with no dependence on the cost specification, any behavioural elasticity, or any simulated re-optimisation. Raising the threshold to £100,000 leaves the rate unchanged and therefore *relocates* the zone without altering its width: it is the same £21,250 band, transported to the new cutoff. A banded reduced rate *shrinks* the zone, because a falls with τ : at $T^* = £85,000$, lowering the band rate to 15% cuts it to $£85,000 \times 0.15/0.85 = £15,000$, and to 10% cuts it to $£85,000 \times 0.10/0.90 = £9,444$. A graduated taper has *no* notch and hence no dominated region: marginal net revenue declines smoothly through the band rather than off a cliff, so the band of dominated turnover—and with it the discrete incentive to suppress turnover—is removed entirely. The dominated region is thus the paper’s exact, model-free measure of the distortion the notch imposes, and the analytic statement above is exactly how much of that distortion each reform of Section 4.1 removes.

7 Conclusion

This paper contributes an open, reproducible firm-level microsimulation for analysing the UK VAT registration threshold, released in full—including its synthetic-data generator—for replication. The framework assembles four components: a synthetic firm dataset built from ONS business-structure counts and calibrated to HMRC VAT aggregates; a

Table 3: Summary of headline estimates. This table consolidates results reported and defined in the body sections cross-referenced below; it is a reference summary, not a new analysis. All quantities refer to the £85,000 data-year baseline. Static revenue figures are 2025–26.

Object	Symbol / definition	Value	Source
<i>Reduced-form bunching (£85k data year)</i>			
Bunching ratio (geometry, <i>not</i> an elasticity)	$b = \Delta y^* / y^*$	0.060	§5
Aggregate displaced share	Π	0.211	§5
Bunching-implied turnover elasticity (descriptive)	ε (notch width)	≈ 0.45	§5
<i>Static costing and structural distortion</i>			
Static cost, £85k to £90k anchor	—	$\approx -£175\text{m}$	§4
Static sweep, £90k baseline	per £5k step	$\sim £150\text{--}220\text{m}$	Table 1
Dominated region (misallocation zone)	$a = T^* \tau / (1 - \tau)$	£21,250	§6.2

static costing of threshold-location and schedule-shape reforms; a reduced-form bunching estimator; and a [Kleven and Waseem \(2013\)](#) notch model whose *exact dominated region* characterises the distortion the threshold imposes and how each reform changes it. Its value is as usable policy infrastructure rather than a structural identification result: I make no claim to recover policy-invariant firm primitives, unlike [Garicano et al. \(2016\)](#) and [Gourio and Roys \(2014\)](#), which identify such primitives at a size threshold. The analysis is anchored to the 2023–24 data year, in which the statutory threshold was £85,000. Table 3 collects the headline objects.

Findings. First, *the pipeline reproduces HMRC’s static costing*. My static costing of the real £85,000 to £90,000 threshold move agrees with HMRC’s published figures to within £1m in 2025–26, and a threshold sweep (Table 1) shows that moving the threshold is approximately linear in revenue and adequately costed by a static calculation. This is the validation anchor for the rest of the framework.

Second, *the behavioural response at the notch is modest*. Firms bunch below the threshold—a bunching ratio $b = 0.060$, which once normalised to the [Liu et al. \(2021\)](#) excess-bunching statistic is of the same order as their estimate. The bunching-implied turnover elasticity (≈ 0.45 from the notch width) is modest, but I report it descriptively rather than as a structural estimate: [Liu et al. \(2021\)](#) decline to convert their bunching mass into an elasticity at a notch, and a turnover notch confounds an output-supply response with a demand-side one, so the number is an upper envelope rather than a welfare-sufficient statistic.

Third, *the structural notch model characterises the distortion exactly*. The notch creates a dominated region of width $a = T^* \tau / (1 - \tau) = £21,250$, a pure misallocation zone in which no firm optimally locates. This object is exact: it depends on τ and T^* alone, independent of the cost specification, the returns-to-scale parameter α , and any smoothing parameter, and it is the structural counterpart of the missing mass measured just above

the threshold. It is the clean, robust index of the distortion the threshold imposes.

Fourth, *I cost a menu of schedule reforms statically and state exactly how each changes that distortion*. Re-applying each candidate schedule to the fixed turnover distribution (Section 4.1) gives static revenue effects of –£374m for raising the threshold to £100,000, –£335m for a graduated taper over [£85k, £105k], and –£357m and –£178m for 10% and 15% banded reduced rates. The dominated region responds analytically: raising the threshold relocates the £21,250 zone unchanged, a 10% band shrinks it to £9,444, a 15% band to £15,000, and a graduated taper removes it altogether. These are the two robust objects—the static cost and the exact change in the misallocation zone—that I report for each reform.

Limitations. The model is a *turnover-tax-notch approximation*, and this is its first-order limitation. It treats VAT as a cost on turnover, whereas real VAT is levied on value added with input reclaim, and roughly half of below-threshold firms register *voluntarily* because reclaim outweighs the output liability—the central fact in [Liu et al. \(2021\)](#). The effective notch is therefore smaller (and, for net-reclaim firms, can reverse sign) than my flat treatment implies. The microdata are *synthetic*—ONS business-structure counts calibrated to HMRC aggregates—so agreement with HMRC totals is a consistency check, not external validation. The analysis is *partial-equilibrium*, abstracting from VAT pass-through into prices ([Benedek et al., 2015](#); [Bellon et al., 2026](#)). And while the dominated-region distortion measure is exact and analytic, any *behavioural* revenue or distributional magnitudes from re-solving firm behaviour would be parameter-sensitive (to the smoothing parameter k and the re-optimisation window), which is why I headline the static costs and the analytic distortion measure rather than re-optimised numbers.

Future work. The static reform costs and the analytic dominated-region results are robust, but two natural extensions remain. First, a fuller *behavioural* treatment that headlines re-optimised revenue and distributional magnitudes from re-solving firm behaviour under each schedule. Such an exercise requires calibrating firm-level cost parameters and a turnover elasticity, and the resulting magnitudes are sensitive to the smoothing parameter k and the re-optimisation window, so I leave a disciplined treatment to future work. Second, a full *welfare/MVPPF analysis* that models value added and input reclaim would convert the dominated-region distortion measure into the cost-benefit terms policy needs, and support the design of *revenue-neutral* schedule variants that neutralise the notch at no net fiscal cost. Modelling *voluntary registration* and the input-VAT margin explicitly would let the framework represent the full UK VAT population rather than the consumer-facing subpopulation it currently best describes. Allowing sector heterogeneity and linking the microdata to [PolicyEngine \(2025\)](#) would refine both the bunching estimates and any counterfactuals. Above all, *validation against administrative firm-level data* would replace the present internal-consistency checks with genuine out-of-sample tests.

References

- Almunia, Miguel and David Lopez-Rodriguez (2018) “Under the Radar: The Effects of Monitoring Firms on Tax Compliance,” *American Economic Journal: Economic Policy*, 10 (1), 1–38.
- Asatryan, Zareh and Andreas Peichl (2017) “Responses of Firms to Tax, Administrative and Accounting Rules: Evidence from Armenia,” *CESifo Working Paper No. 6754 / ZEW Discussion Paper No. 16-065*.
- Bashir, Muhammad, Zehra Farooq, Usama Jamal, and Mazhar Waseem (2024) “Size-Based Policies and Firm Growth: Evidence from Pakistan,” *Working Paper*.
- Bellon, Matthieu, Alexander Copestake, and Wenzhang Zhang (2026) “Supply and Demand Determinants of Heterogeneous VAT Pass-Through,” *National Tax Journal (forthcoming)*.
- Benedek, Dora, Ruud A. de Mooij, Michael Keen, and Philippe Wingender (2015) “Estimating VAT Pass Through,” *IMF Working Paper (WP/15/214)*.
- Best, Michael Carlos, Anne Brockmeyer, Henrik J. Kleven, Johannes Spinnewijn, and Mazhar Waseem (2015) “Production versus Revenue Efficiency with Limited Tax Capacity: Theory and Evidence from Pakistan,” *Journal of Political Economy*, 123 (6), 1311–1355.
- Chetty, Raj, John N. Friedman, Tore Olsen, and Luigi Pistaferri (2011) “Adjustment Costs, Firm Responses, and Micro vs. Macro Labor Supply Elasticities,” *Quarterly Journal of Economics*, 126 (2), 749–804.
- Garicano, Luis, Claire Lelarge, and John Van Reenen (2016) “Firm Size Distortions and the Productivity Distribution: Evidence from France,” *American Economic Review*, 106 (11), 3439–3479.
- Gourio, François and Nicolas Roys (2014) “Size-Dependent Regulations, Firm Size Distribution, and Reallocation,” *Quantitative Economics*, 5 (2), 377–416.
- Harju, Jarkko, Tuomas Matikka, and Timo Rauhanen (2019) “Compliance Costs vs. Tax Incentives: Why Do Entrepreneurs Respond to Size-Based Regulations?” *Journal of Public Economics*, 173, 139–164.
- HM Revenue and Customs (2024) “VAT Registration Thresholds,” GOV.UK guidance, <https://www.gov.uk/vat-registration-thresholds>.
- Kleven, Henrik J. (2016) “Bunching,” *Annual Review of Economics*, 8, 435–464.
- Kleven, Henrik J. and Mazhar Waseem (2013) “Using Notches to Uncover Optimization Frictions and Structural Elasticities: Theory and Evidence from Pakistan,” *Quarterly Journal of Economics*, 128 (2), 669–723.
- Liu, Li, Ben Lockwood, Miguel Almunia, and Eddy H. F. Tam (2021) “VAT Notches, Voluntary Registration, and Bunching: Theory and UK Evidence,” *Review of Economics and Statistics*, 103 (1), 151–164.

- Liu, Li, Ben Lockwood, and Eddy H. F. Tam (2024) “Small Firm Growth and the VAT Threshold: Evidence for the UK,” *IMF Working Paper No. 2024/033*.
- Nandi, Tushar and Ross Warwick (2020) “How Businesses Respond to VAT: Evidence from West Bengal, India,” *TaxDev / Institute for Fiscal Studies*.
- Office for Budget Responsibility (2023) “The Impact of the Frozen VAT Registration Threshold,” *Economic and Fiscal Outlook*, March 2023, <https://obr.uk/box/the-impact-of-the-frozen-vat-registration-threshold/>.
- Onji, Kazuki (2009) “The Response of Firms to Eligibility Thresholds: Evidence from the Japanese Value-Added Tax,” *Journal of Public Economics*, 93 (5–6), 766–775.
- PolicyEngine (2025) “PolicyEngine UK: Open-Source Tax-Benefit Microsimulation,” <https://policyengine.org>.
- Saez, Emmanuel (2010) “Do Taxpayers Bunch at Kink Points?” *American Economic Journal: Economic Policy*, 2 (3), 180–212.
- Seely, Antony (2024) “VAT: The Registration Threshold,” *House of Commons Library Briefing Paper*.

A Appendix

This appendix collects supporting material in two parts: data construction and calibration robustness (Appendix A.1); and statistical inference for the reduced-form bunching estimates (Appendix A.2).

A.1 Data construction detail

Section 3 describes the two-stage synthetic-population pipeline (`analysis/gen_full_param.py`, parameterised by `VAT_THRESHOLD`) and its calibration to HMRC and ONS aggregates. This appendix records the implementation detail deferred from that account.

Each of the approximately 2.94 million firm rows carries the fields `annual_turnover_k`, `annual_input_k`, `weight`, and `vat_registered`; calibration weights are chosen so the weighted totals reproduce the official targets while the population sums to about 2.0 million firms. Because the population is calibrated *to* the HMRC aggregates, agreement between my weighted outputs and those aggregates is a check of *internal consistency*, not external validation: any target used in calibration is reproduced by construction. I therefore read the $\sim 87\%$ calibration accuracy, and the close match of the static £85,000 $\rightarrow$ £90,000 costing to HMRC’s published figures, as consistency checks on the pipeline rather than out-of-sample tests of firm behaviour.

Counterfactual exclusion window and polynomial degree. The no-VAT counterfactual density of Section 5 is fitted by polynomial regression on the observed density outside a manipulation window around the threshold. The baseline excludes $[T^* - 15k, T^* + 15k]$ and fits a cubic. Widening the window guards against contamination of the counterfactual by the behavioural response but reduces the data available for the fit; raising the degree improves in-sample fit but risks over-fitting the tails. The sensitivity of the excess-mass and bunching-ratio estimates to window width and degree, with associated uncertainty, is reported in Appendix A.2.

Subsample for dynamic plots. The dynamic re-optimisation figures of the companion structural step are generated on a 20% random subsample of the firm rows, to keep the firm-by-firm FOC inversion and re-resolution tractable, while the static revenue sweep and the calibration use the full dataset. The 20% subsample reproduces the full-dataset density step and bunching mass to within sampling error, so subsampling affects only the smoothness of the plotted curves, not the substantive conclusions.

A.2 Inference for the bunching estimator

This appendix documents the inference procedure for the reduced-form bunching estimates of Section 5: the mass-conservation constraint that disciplines the counterfactual density, the bootstrap that delivers standard errors, and the sensitivity of the estimates to the polynomial degree, the exclusion window, the sigmoid steepness k , and the effective wedge τ_e . All numbers reported here and in the calibration step are produced by

the script `analysis/bunching_inference.py`, run on the £85,000 (2023–24) weighted synthetic firm data; nothing is hardcoded as a result.

Mass-conservation constraint

The canonical bunching estimator of [Saez \(2010\)](#), [Kleven and Waseem \(2013\)](#) and [Chetty et al. \(2011\)](#) imposes that the counterfactual density conserves mass: every firm removed from above the threshold reappears as excess mass below it. An unconstrained fit by plain polynomial OLS on bins outside a hand-set window computes the excess mass E (below T^*) and the missing mass Δ_R (above T^*) *independently*, with no guarantee that $E = \Delta_R$ and no endogenous location of the marginal buncher.

I impose the constraint in two ways. First, the counterfactual polynomial in $(y - T^*)$ is fit by an iterated weighted least-squares procedure in which the density of the bins above the excluded region is rescaled until the fitted counterfactual integrates to the *observed* total mass over the estimation range [£20k, £140k]. This forces the bunching mass to be reabsorbed into the counterfactual rather than discarded. The regressor is normalised to $[-1, 1]$ for numerical stability of the high-degree Vandermonde basis. Second, the upper edge of the excluded (manipulation) region—the *marginal buncher* y_R —is located *endogenously* by integrating the missing mass upward from T^* until cumulative missing mass above equals cumulative excess mass below,

$$\int_{T^*}^{y_R} [f^{\text{cf}}(y) - f^{\text{obs}}(y)]_+ dy = \int_{T^*-W}^{T^*} [f^{\text{obs}}(y) - f^{\text{cf}}(y)]_+ dy = E,$$

rather than being fixed at an arbitrary window edge. The excess turnover span of the marginal buncher, $\Delta y^* = y_R - T^*$, is then used by the notch-width elasticity (Method B).

Imposing the constraint matters because the unconstrained fit grossly violates adding-up: it delivers $E = 8,179$ but $\Delta_R = 17,792$, whereas the constrained estimator returns $E = 8,712$ and $\Delta_R = 9,938$, which agree to within sampling noise. The accompanying movement in the headline estimates (b from 0.054 to 0.060, σ from 4.33 to 4.85) is reported in the calibration step.

Bootstrap standard errors

I compute standard errors by a firm-level bootstrap. In each of $B = 200$ replications I draw n firms with replacement from the estimation sample, with draw probabilities proportional to the survey weight, re-bin the resampled firms into £1,000 bins, and re-run the entire estimator—counterfactual fit, marginal-buncher location, and all downstream statistics—on the replicate. The standard error of each statistic is the standard deviation of its bootstrap distribution, and the 95% confidence interval is the 2.5th–97.5th percentile interval. Resampling firms (rather than residuals) propagates both sampling variation and the variation induced by re-binning. Table [A.1](#) reports the results.

Table A.1: Bootstrap inference, £85,000 data ($B = 200$ firm resamples).

Statistic	Point	Std. error	95% CI
Bunching ratio b	0.060	0.005	[0.050, 0.068]
Excess mass E	8,712	662	[7,877, 10,483]
Substitution elasticity σ	4.85	0.12	[4.61, 5.10]
Displaced share Π	0.211	0.005	[0.202, 0.221]
Notch-width elasticity (median)	0.448	0.027	[0.419, 0.519]
Marginal-buncher elasticity	0.657	0.082	[0.536, 0.828]

Table A.2: Degree $\times$ window sensitivity (point estimates), £85,000 data. Cells with $b < 0$ leave σ undefined.

Degree	Window	b	E	σ	Notch-width (median)
6	£10k	0.079	7,719	5.69	0.42
6	£15k	0.046	6,900	5.24	0.45
6	£20k	0.017	4,101	4.44	0.38
7	£10k	0.095	9,073	5.32	0.47
7	£15k	0.060	8,712	4.85	0.45
7	£20k	0.036	7,785	4.89	0.45

Sensitivity to polynomial degree and exclusion window

Table A.2 maps the point estimates across a grid of polynomial degree $\in \{5, 6, 7, 8\}$ and symmetric exclusion window $\in \{\text{£}10\text{k}, \text{£}15\text{k}, \text{£}20\text{k}, \text{£}25\text{k}\}$. The estimates are stable across the central cells (degree 6–7, window £10–£20k): the substitution elasticity stays in roughly 4.4–5.7 and the excess mass in 4,000–9,000. The corner cells are less reliable: at very wide windows (£25k) the excluded region begins to swallow genuine curvature in the density, the fitted counterfactual crosses the observed density, and b can turn slightly negative (so σ is undefined); at the lowest degree (5) the inflexible trend over-fits the bunching mass into the counterfactual and inflates b . I therefore report the degree-7, £15k-window specification as the baseline, with the bootstrap CIs above spanning the variation across the stable interior of the grid.

Sensitivity to the sigmoid steepness

The notch-width elasticity uses the effective wedge $\tau_{\text{eff}} = \tau_{\text{max}}/2$, which holds at the threshold for *any* sigmoid steepness k , so the data-side elasticity is k -invariant. The steepness k instead governs the *structural* model: it sets the width of the output-dominated (“forbidden”) band of the smoothed schedule, defined as the turnover region in which marginal net revenue $(1 - \tau(y)) - y\tau'(y)$ is negative. My code computes this width directly and compares it to the exact Kleven and Waseem (2013) dominated region for a 20% notch at £85,000, $T^*\tau/(1 - \tau) = \text{£}21.25\text{k}$. The width is strongly k -dependent: $k = 0.0001$ leaves essentially the entire local band dominated (a near-kink), $k = 0.001$ gives £19.7k

(closest to the exact £21.25k), $k = 0.002$ gives £10.6k, $k = 0.005$ gives £4.6k, and $k = 0.01$ gives only £2.5k; the empirical excess span of the marginal buncher is $\Delta y^* = £5.4k$. This is the basis for the choice of $k = 0.001$ in the calibration step: its implied dominated region best matches both the exact notch region and the order of magnitude of the empirical span, whereas $k = 0.01$ understates the band by roughly an order of magnitude. I report results at $k = 0.001$ and treat the spread above as a calibration sensitivity rather than a point identification of k .

Sensitivity to the effective wedge

The CES substitution elasticity is recovered as $\sigma = \ln(RR) / \ln(1 + \tau_e)$, where RR is the ratio of counterfactual-to-observed above/below mass ratios. Because the wedge enters only through $\ln(1 + \tau_e)$, σ scales roughly as $1 / \ln(1 + \tau_e)$ while the displaced share $\Pi = 1 - (1 + \tau_e)^{-\sigma}$ is essentially invariant (it is pinned by RR). On my data, $\tau_e = 0.025$ gives $\sigma = 9.61$, $\tau_e = 0.05$ gives $\sigma = 4.85$, $\tau_e = 0.075$ gives $\sigma = 3.27$, and $\tau_e = 0.10$ gives $\sigma = 2.48$, while $\Pi = 0.211$ throughout. The magnitude of σ is therefore conditional on τ_e : I adopt $\tau_e = 0.05$ as a baseline effective registration wedge and report Π , which is robust to it, as the policy-relevant displacement object.